

State density II: the leading coefficient for a unique minimal coordinate

$$n^{p/2} Z(n) \rightarrow \frac{1}{k_0} \int_{(0,b]^d} v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(0, v)) \eta(0, v) dv$$

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Abstract

The first formalised “state density” coefficient. With coordinates (u, v) , $u \in (0, b]$ the unique coordinate attaining the minimal candidate exponent $p = (h_0 + 1)/k_0$ and $v \in (0, b]^d$ the noncritical ones ($q_i > p$), for ξ, η jointly continuous and Lipschitz at $u = 0$ uniformly in v ,

$$\begin{aligned} n^{p/2} \int_v v^{h'} \int_0^b u^{h_0} \eta(u, v) e^{-\beta n (u^{k_0} v^{k'})^2 + \beta \sqrt{n} u^{k_0} v^{k'} \xi(u, v)} du dv \\ \rightarrow \frac{1}{k_0} \int_v v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(0, v)) \eta(0, v) dv. \end{aligned}$$

The leading coefficient is the fluctuation mass $A_{p-1}(\xi(0, v))$, weighted by $\eta(0, v)$, integrated along the exceptional divisor $\{u = 0\}$ against the density $v^{h'-k'p}$, integrable exactly because the other candidate exponents are strictly larger. Zero ~~sorry~~/axiom.

1 Setting

For fixed v the inner integral is the one-dimensional standard integral at scale $c = \sqrt{n} v^{k'}$ with data $\xi(\cdot, v), \eta(\cdot, v)$; unit 69 gives $c^p I_v(c) \rightarrow \frac{1}{k_0} A_{p-1}(\xi(0, v)) \eta(0, v)$ and a bound uniform in c . Since $n^{p/2} = c^p (v^{k'})^{-p}$, dominated convergence in v with the dominating function $K v^{h'} (v^{k'})^{-p}$ finishes.

2 The things formalised

```
noncomputable def stateIntegral ( b n : ) (h k : ) {d : } (k' h' : Fin d → )
  ( : → (Fin d → ) → ) : :=
  v, (( i, v i ^ h' i) * u in Ioc 0 b, u ^ h * u v
    * Real.exp (- * n * (u ^ k * i, v i ^ k' i) ^ 2
      + * Real.sqrt n * (u ^ k * i, v i ^ k' i) * u v)) (boxMeasure b d)
noncomputable def divisorDensity ... (v : Fin d → ) : :=
  ( i, v i ^ h' i) * ( i, v i ^ k' i) ^ (-((h : ) + 1) / k))
  * (1 / (k : ) * weightedMass ( 0 v) ((h : ) + 1) / k - 1) * 0 v)
theorem stateIntegral_tendsto ...
  (hq : i, ((h : ) + 1) / k < ((h' i : ) + 1) / k' i) ...
  (h : u Ioc 0 b, v, | u v - 0 v| C * u)
  (h : u Ioc 0 b, v, | u v - 0 v| C' * u) :
  Tendsto (fun n => n ^ (((h : ) + 1) / k / 2) * stateIntegral b n h k k' h' )
    atTop ( ( v, divisorDensity h k k' h' v (boxMeasure b d)))
```

3 Proof strategy

`tendsto_integral_filter_of_dominated_convergence` over the box measure with $F_n(v) = n^{p/2} v^{h'} I_v(\sqrt{n} v^{k'})$. Measurability of F_n is Fubini's `StronglyMeasurable.integral_prod_right'` applied to the jointly continuous integrand. The bound uses unit 69's uniform estimate together with $n^{p/2} = c^p (v^{k'})^{-p}$ and the factorisation $v^{h'} (v^{k'})^{-p} = \prod_i v_i^{h'_i} (v_i^{k'_i})^{-p}$ (`Real.finsetProd_rpow`); the dominating product is integrable by `Integrable.fintype_prod`, each factor being $v_i^{h'_i - k'_i p}$ with exponent > -1 (`intervalIntegrable_rpow'`). The pointwise limit is unit 69's theorem composed with $c = \sqrt{n} v^{k'} \rightarrow \infty$. Bounds L, M for ξ, η come from continuity on the compact closed box.

4 Roadblocks and resolutions

- In a nested integral the trailing attaches to the innermost ; parenthesise the inner integral.
- A binder `i, f i * g` swallows `g`; parenthesise the product before multiplying.
- `Integrable.fintype_prod` lives in `Mathlib.MeasureTheory.Integral.Pi`, outside the import closure; `Real.finset_prod_rpow` is deprecated for `Real.finsetProd_rpow`, stated in the opposite direction.

5 What was Mathlib, what was new

Mathlib: `tendsto_integral_filter_of_dominated_convergence`,
`StronglyMeasurable.integral_prod_right'`, `Integrable.fintype_prod`,
`Real.finsetProd_rpow`, `IsCompact.exists_bound_of_continuousOn`. New: the state integral, the divisor density, and the leading coefficient theorem.

6 Lessons

The state-density coefficient is reached without any Mellin transform: once the one-dimensional theory is available at a general scale with a uniform bound, dominated convergence along the divisor does the rest. The integrability of the divisor density is exactly the “strictly larger exponents” condition.

7 Outcome

For a unique minimal coordinate the leading coefficient of `thm:TaylorTree` is formalised for general (Lipschitz) ξ, η : $C = \frac{1}{k_0} \int v^{h' - k'p} A_{p-1}(\xi(0, v)) \eta(0, v) dv$. Next: positivity and the asymptotic equivalence form, the Fubini bridge to the $(d + 1)$ -dimensional box, and then several minimal coordinates (the logarithmic case).